Mini 2 Lab 1

Activating tensorflow python3 environment and installing flask

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\STUDENT> cd .\thuotfl\Scripts\
PS C:\Users\STUDENT\thuotfl\Scripts> .\Activate.ps1
(thuotfl) PS C:\Users\STUDENT\thuotfl\Scripts> pip3 install flask
Collecting flask
  Downloading flask-3.1.0-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: Werkzeug>=3.1 in c:\users\student\thuotfl\lib\site-packages (from flask) (3.1.3)
Collecting Jinja2>=3.1.2 (from flask)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.2 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting blinker>=1.9 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting colorama (from click>=8.1.3->flask)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\student\thuotfl\lib\site-packages (from Jinja2>=3.1.2->flask) (3.0.2)
Downloading flask-3.1.0-py3-none-any.whl (102 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.1.8-py3-none-any.whl (98 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: Jinja2, itsdangerous, colorama, blinker, click, flask
Successfully installed Jinja2-3.1.6 blinker-1.9.0 click-8.1.8 colorama-0.4.6 flask-3.1.0 itsdangerous-2.2.0

[notice] A new release of pip is available: 25.0 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(thuotfl) PS C:\Users\STUDENT\thuotfl\Scripts>
```
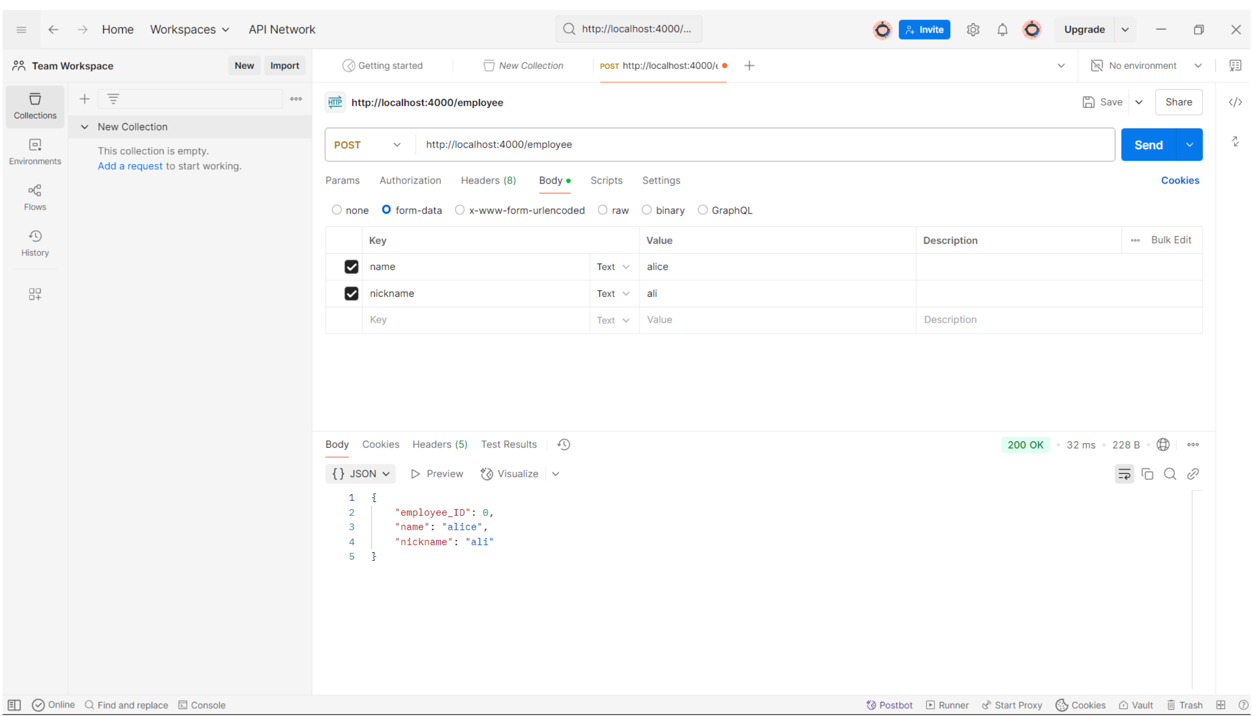
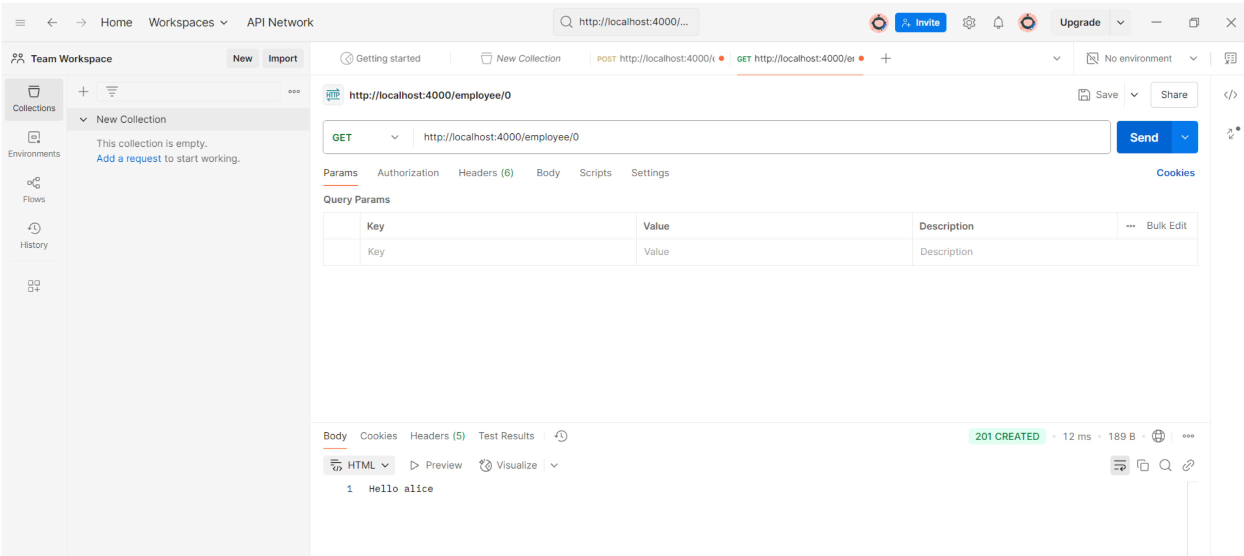

Testing postman

Post method

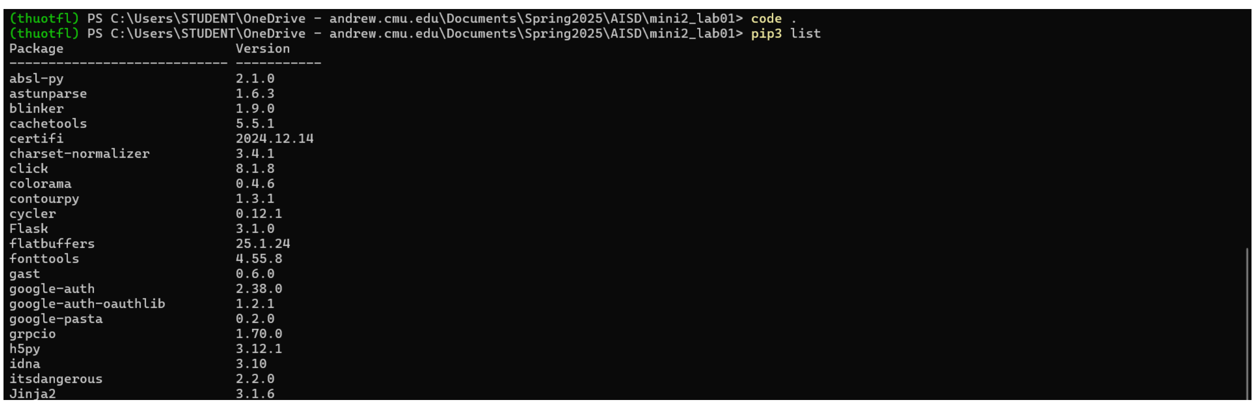


Get method

`http://localhost:4000/employee/0` (Using the ID from the previous step, here it's `0`)



Using pip3 list to confirm all libraries are present

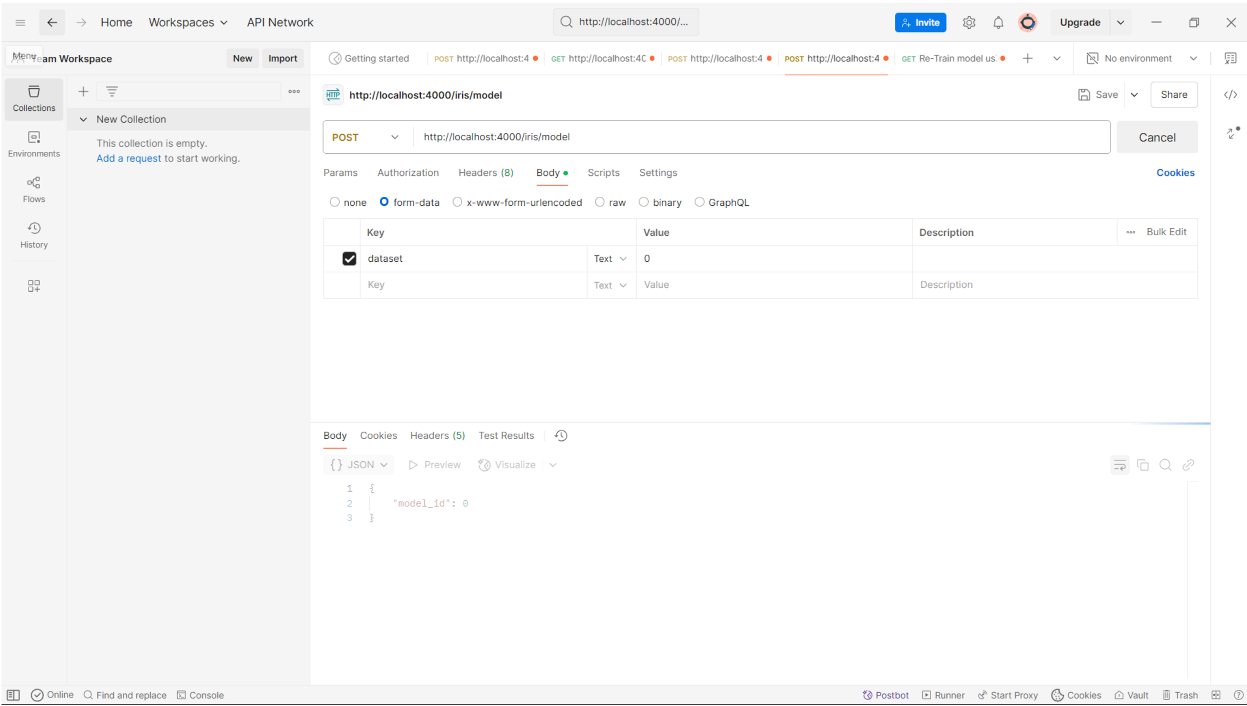


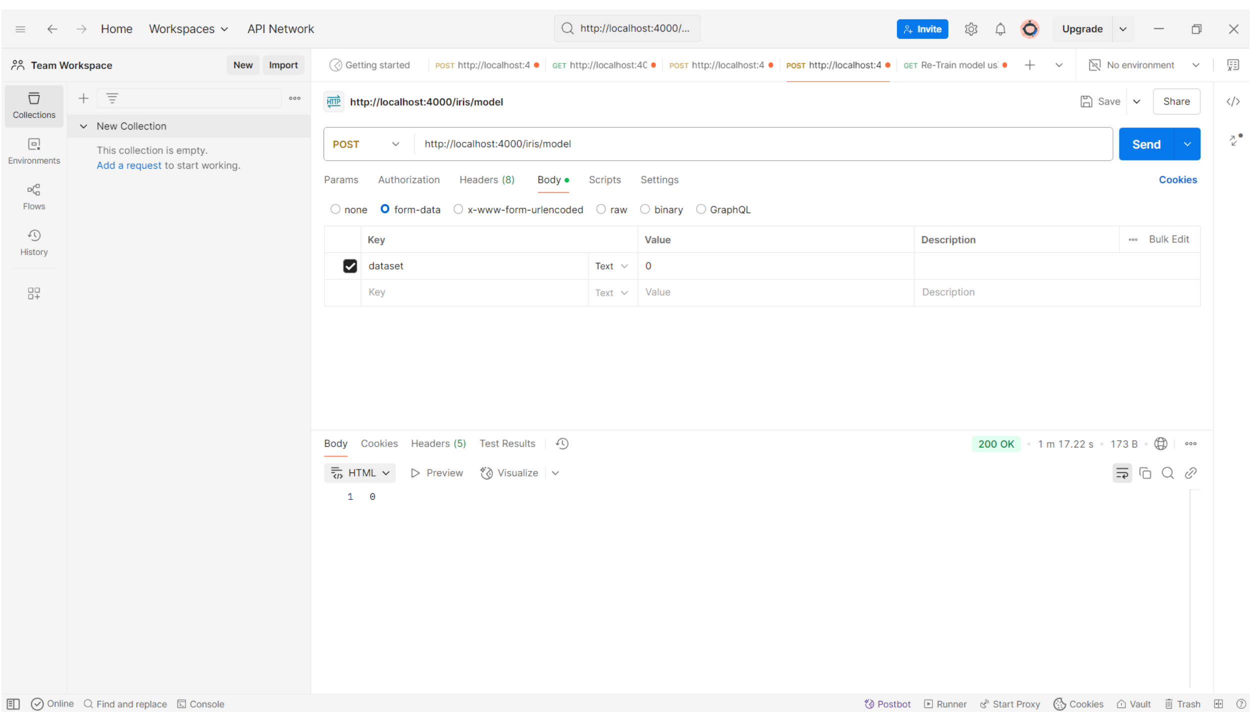
Prep: Installing and running the simple Iris classification model

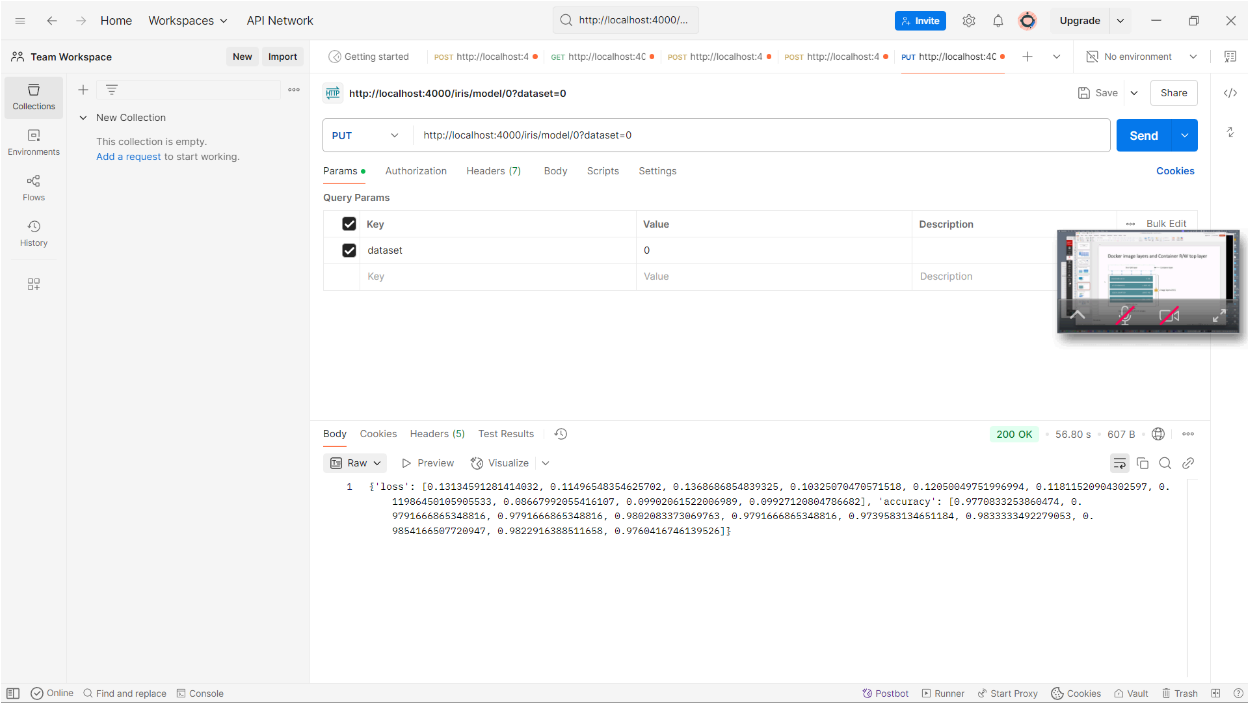
```
(thuotfl) PS C:\Users\STUDENT\OneDrive - andrew.cmu.edu\Documents\Spring2025\AISD\mini2_lab01> python
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> from base_iris_lab1 import load_local, build, train, score
2025-03-20 11:37:17.507847: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on. You may see slightly different numerical results due to flo
ating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING:tensorflow:From C:\Users\STUDENT\thuotfl\Lib\site-packages\keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated.
Please use tf.compat.v1.losses.sparse_softmax_cross_entropy instead.

starting up iris model service
>>> dataset = load_local()
load local default data
>>> print(dataset)
0
>>> model = build()
WARNING:tensorflow:From C:\Users\STUDENT\thuotfl\Lib\site-packages\keras\src\backend.py:873: The name tf.get_default_graph is deprecated. Please use tf.comp
at.v1.get_default_graph instead.

2025-03-20 11:40:45.435161: I tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow binary is optimized to use available CPU instructions in pe
rformance-critical operations.
To enable the following instructions: SSE SSE2 SSE3 SSE4.1 SSE4.2 AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler f
lags.
WARNING:tensorflow:From C:\Users\STUDENT\thuotfl\Lib\site-packages\keras\src\optimizers\__init__.py:309: The name tf.train.Optimizer is deprecated. Please u
se tf.compat.v1.train.Optimizer instead.

>>> print(model)
0
>>> hist = train(model,dataset)
Epoch 1/10
WARNING:tensorflow:From C:\Users\STUDENT\thuotfl\Lib\site-packages\keras\src\utils\tf_utils.py:492: The name tf.ragged.RaggedTensorValue is deprecated. Plea
se use tf.compat.v1.ragged.RaggedTensorValue instead.

WARNING:tensorflow:From C:\Users\STUDENT\thuotfl\Lib\site-packages\keras\src\engine\base_layer_utils.py:384: The name tf.executing_eagerly_outside_functions
 is deprecated. Please use tf.compat.v1.executing_eagerly_outside_functions instead.

  1/119 [..............................] - ETA: 4:05 - loss: 5.0834 - accura 17/119 [===>..........................] - ETA: 0s - loss: 2.7057 - accuracy 33/
119 [=======>......................] - ETA: 0s - loss: 2.5773 - accuracy 59/119 [=============>................] - ETA: 0s - loss: 1.9720 - accuracy 82/119
[===================>..........] - ETA: 0s - loss: 1.7518 - accuracy107/119 [=========================>....] - ETA: 0s - loss: 1.5935 - accuracy119/119 [===
===========================] - 2s 2ms/step - loss: 1.5234 - accuracy: 0.3782
Epoch 2/10
  1/119 [..............................] - ETA: 0s - loss: 0.6677 - accuracy 30/119 [======>.......................] - ETA: 0s - loss: 0.9214 - accuracy 52/
119 [============>.................] - ETA: 0s - loss: 0.9021 - accuracy 72/119 [=================>............] - ETA: 0s - loss: 0.8739 - accuracy 94/119
[======================>.......] - ETA: 0s - loss: 0.8482 - accuracy114/119 [===========================>..] - ETA: 0s - loss: 0.8362 - accuracy119/119 [===
===========================] - 0s 2ms/step - loss: 0.8267 - accuracy: 0.6555
```

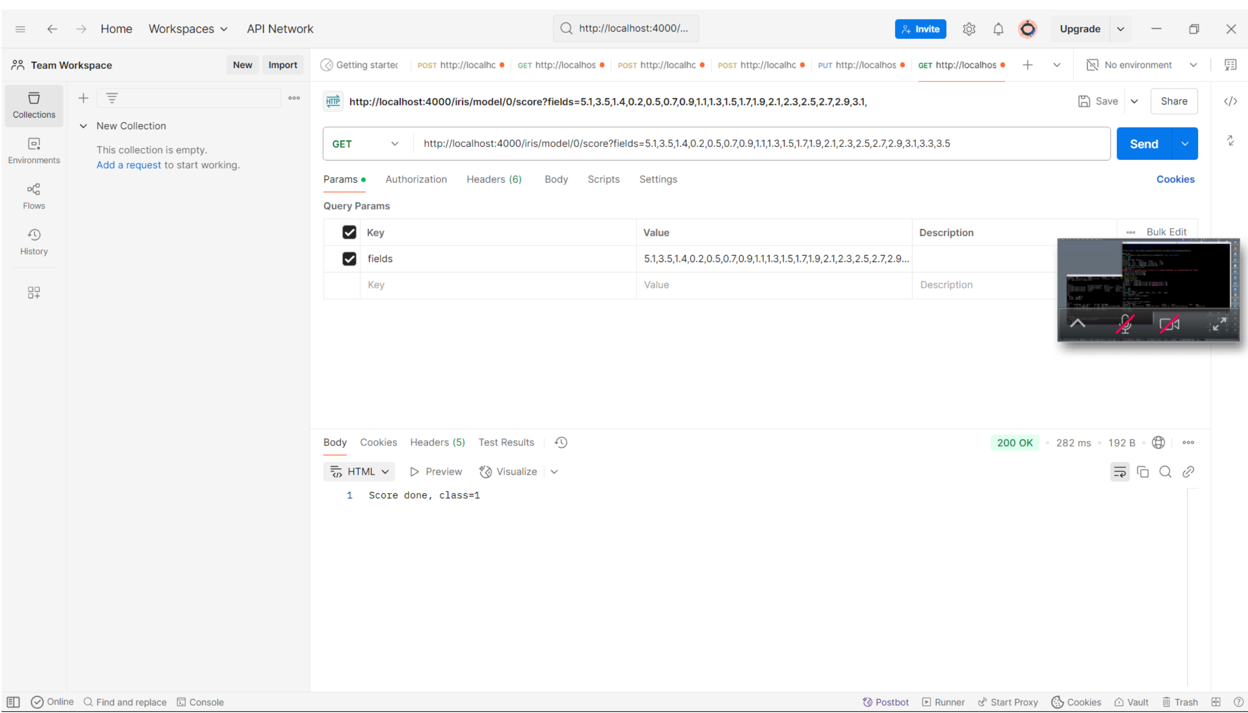
```
Windows PowerShell
Epoch 7/10
  1/119 [..............................] - ETA: 0s - loss: 0.6277 - accuracy 25/119 [=====>........................] - ETA: 0s - loss: 0.4906 - accuracy 55/
119 [============>.................] - ETA: 0s - loss: 0.4560 - accuracy 81/119 [===================>..........] - ETA: 0s - loss: 0.4433 - accuracy104/119
[=========================>....] - ETA: 0s - loss: 0.4233 - accuracy119/119 [==============================] - 0s 2ms/step - loss: 0.4167 - accuracy: 0.8739
Epoch 8/10
  1/119 [..............................] - ETA: 0s - loss: 0.0266 - accuracy 26/119 [=====>........................] - ETA: 0s - loss: 0.3812 - accuracy 49/
119 [===========>..................] - ETA: 0s - loss: 0.3789 - accuracy 72/119 [=================>............] - ETA: 0s - loss: 0.3390 - accuracy 92/119
[======================>.......] - ETA: 0s - loss: 0.3760 - accuracy111/119 [==========================>...] - ETA: 0s - loss: 0.3872 - accuracy119/119 [===
===========================] - 0s 2ms/step - loss: 0.3877 - accuracy: 0.8319
Epoch 9/10
  1/119 [..............................] - ETA: 0s - loss: 0.4898 - accuracy 22/119 [====>.........................] - ETA: 0s - loss: 0.2573 - accuracy 40/
119 [=========>....................] - ETA: 0s - loss: 0.2969 - accuracy 63/119 [==============>...............] - ETA: 0s - loss: 0.3298 - accuracy 84/119
[====================>.........] - ETA: 0s - loss: 0.3609 - accuracy107/119 [=========================>....] - ETA: 0s - loss: 0.3719 - accuracy119/119 [===
===========================] - 0s 2ms/step - loss: 0.3681 - accuracy: 0.9076
Epoch 10/10
  1/119 [..............................] - ETA: 0s - loss: 0.5245 - accuracy 20/119 [====>.........................] - ETA: 0s - loss: 0.3586 - accuracy 39/
119 [========>.....................] - ETA: 0s - loss: 0.3516 - accuracy 60/119 [==============>...............] - ETA: 0s - loss: 0.3566 - accuracy 81/119
[===================>..........] - ETA: 0s - loss: 0.3579 - accuracy104/119 [=========================>....] - ETA: 0s - loss: 0.3520 - accuracy119/119 [===
===========================] - 0s 2ms/step - loss: 0.3484 - accuracy: 0.9328
{'loss': [1.5233547687530518, 0.8266677856445312, 0.6463104486465454, 0.5420878529548645, 0.48479241132736206, 0.44516465067863464, 0.41668787598609924, 0.3
8770243525505066, 0.3681012988090515, 0.3484492003917694], 'accuracy': [0.3781512677669525, 0.6554622054100037, 0.6722689270973206, 0.7394958138465881, 0.70
58823704719543, 0.8235294222831726, 0.8739495873451233, 0.831932783126831, 0.9075630307197571, 0.9327731132507324]}
Test loss: 0.30222633481025696
Test accuracy: 1.0
1/1 [==============================] - 0s 158ms/step
Actual: [2 2 1 1 0 2 2 1 2 1 0 0 1 0 0 2 2 1 0 0 0 0 1 0 1 1 1 0 0 2]
Predicted: [2 2 1 1 0 2 2 1 2 1 0 0 1 0 0 2 2 1 0 0 0 0 1 0 1 1 1 0 0 2]
Confusion matrix on test data is [[12  0  0]
 [ 0 10  0]
 [ 0  0  8]]
Precision Score on test data is [1. 1. 1.]
Recall Score on test data is [1. 1. 1.]
>>> score(model)
1/1 [==============================] - 0s 43ms/step
[[9.9218315e-01 6.9416249e-03 8.7522902e-04]]
0
'Score done, class=0'
```

```
>>> score(model, SLen=5.1, SWidth=3.5, PLen=1.4, PWidth=0.2)
1/1 [==============================] - 0s 30ms/step
[[9.9361056e-01 5.6323018e-03 7.5706193e-04]]
0
'Score done, class=0'
>>> score(model, SLen=2.1, SWidth=1.5, PLen=0.4, PWidth=0.9)
1/1 [==============================] - 0s 42ms/step
[[0.89664835 0.07678244 0.02656926]]
0
'Score done, class=0'
>>> score(model, SLen=6.1, SWidth=4.5, PLen=1.4, PWidth=2.9)
1/1 [==============================] - 0s 30ms/step
[[9.9872154e-01 1.1842365e-03 9.4223622e-05]]
0
'Score done, class=0'
>>> |
```






Home
Workspaces
API Network
http://localhost:4000/...
Invite
Upgrade
Team Workspace
New
Import
Collections
Environments
Flows
History
New Collection
This collection is empty.
Add a request to start working.
http://localhost:4000/iris/model/0?dataset=0
Save
Share
PUT
http://localhost:4000/iris/model/0?dataset=0
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
dataset
0
Body
Cookies
Headers (5)
Test Results
200 OK
56.80 s
607 B
Raw
Preview
Visualize
1 {'loss': [0.13134591281414032, 0.11496548354625702, 0.1368686854839325, 0.10325070470571518, 0.12050049751996994, 0.11811520904302597, 0.11986450105905533, 0.08667992055416107, 0.09902061522006989, 0.09927120804786682], 'accuracy': [0.9770833253860474, 0.9791666865348816, 0.9791666865348816, 0.9802083373069763, 0.9791666865348816, 0.9739583134651184, 0.9833333492279053, 0.9854166507720947, 0.9822916388511658, 0.9760416746139526]}
Online
Find and replace
Console
Postbot
Runner
Start Proxy
Cookies
Vault
Trash


Home
Workspaces
API Network
http://localhost:4000/...
Invite
Upgrade
Team Workspace
New
Import
Collections
Environments
Flows
History
New Collection
This collection is empty.
Add a request to start working.
http://localhost:4000/iris/model/0/score?fields=5.1,3.5,1.4,0.2,0.5,0.7,0.9,1.1,1.3,1.5,1.7,1.9,2.1,2.3,2.5,2.7,2.9,3.1,
Save
Share
GET
http://localhost:4000/iris/model/0/score?fields=5.1,3.5,1.4,0.2,0.5,0.7,0.9,1.1,1.3,1.5,1.7,1.9,2.1,2.3,2.5,2.7,2.9,3.1,3.3,3.5
Send
Params
Authorization
Headers (6)
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
fields
5.1,3.5,1.4,0.2,0.5,0.7,0.9,1.1,1.3,1.5,1.7,1.9,2.1,2.3,2.5,2.7,2.9...
Body
Cookies
Headers (5)
Test Results
200 OK
282 ms
192 B
HTML
Preview
Visualize
1 Score done, class=1
Online
Find and replace
Console
Postbot
Runner
Start Proxy
Cookies
Vault
Trash

HTTP

HTTP (Hypertext Transfer Protocol) is the foundation of data exchange on the Web, enabling clients, such as web browsers, to request and receive resources like HTML documents, images, and videos from servers. As a client-server protocol, HTTP relies on requests and responses to facilitate communication between these entities. Initially designed in the early 1990s, HTTP has evolved into an extensible protocol that operates over TCP or encrypted TLS connections, supporting both simple document retrieval and more complex interactions like form submissions and dynamic content updates.

A key feature of HTTP is its stateless nature, meaning each request is independent, though mechanisms like cookies enable session-based interactions. The protocol functions at the application layer, relying on underlying network and transport layers to ensure reliability. Between clients and servers, proxies play various roles, such as caching for efficiency, filtering content, balancing loads, and enabling authentication. Additionally, HTTP/2 introduced multiplexing, improving performance by allowing multiple requests to be sent over a single connection.

Despite its simplicity, HTTP provides significant control over caching, authentication, security policies, and session management. Modern advancements continue to refine HTTP's efficiency, with protocols like QUIC aiming to enhance speed and reliability. As the backbone of the internet, HTTP remains indispensable, supporting seamless and scalable web interactions while adapting to new technological demands.

URL

A URL (Uniform Resource Locator) is the unique address of a resource on the internet, used by browsers to retrieve web pages, images, videos, and other content. It consists of several key components: the scheme (protocol like HTTP or HTTPS), authority (domain and optional port), path (location of the resource on the server), parameters (key-value pairs for additional data), and anchor (bookmark within the resource). URLs can be either absolute (complete addresses) or relative (shortened paths that depend on the context of the current document). Web technologies like HTML, CSS, and JavaScript rely heavily on URLs to link resources and display content.

Well-structured semantic URLs enhance user experience, making web addresses easier to understand, remember, and navigate. They also improve accessibility and search engine optimization (SEO). While URLs are mainly technical, their design impacts usability and clarity for users. Understanding URL structure is essential for efficient web browsing, development, and managing online resources effectively.